

PROFESSIONAL SUMMARY

Software Engineer with a strong track record of taking AI and automation systems from prototype to production and owning them after launch. Built and deployed production-grade LLM pipelines, inference APIs, and backend services on AWS; defined CI/CD workflows, managed deployment infrastructure, and maintained systems through iterative improvements post-release. Comfortable setting practical engineering standards across code quality, testing, and deployment in fast-moving, small-team environments. Equally fluent in ML architecture and backend engineering — the kind of profile that does not hand off at the finish line.

EDUCATION

- Ajou University** Suwon, Republic of Korea
Integrated Master's and Ph.D. Program, Department of AI; GPA: 3.89/4.5 *Sept. 2017 – Feb. 2024*

PROFESSIONAL EXPERIENCE

- DST International Co., Ltd.** Seoul, Republic of Korea
Research Engineer *May 2024 – Present*
 - Production AI Systems — Concept to Ownership:** Designed, built, and maintained a production RAG-based knowledge management system end-to-end: document ingestion, chunking, embedding pipelines, vector storage via Chroma.DB, semantic retrieval, and LLM inference. Owned the system post-launch including bug fixes, performance tuning, and iterative improvements based on real usage.
 - Agentic & LLM Workflow Engineering:** Prototyped and productionised multi-step agentic workflows using LangChain and Ollama, enabling autonomous document reasoning and tool-calling over internal knowledge bases. Iterated on retrieval and reranking strategies using measurable relevance baselines to drive continuous improvement.
 - Engineering Standards & CI/CD Ownership:** Defined and implemented CI/CD pipelines using GitHub Actions, AWS S3, and CodeDeploy for AI-enabled services. Established deployment, testing, and release standards for the team — reducing manual overhead and ensuring systems remained stable and auditable post-launch.
 - Backend API Development & Integration:** Built RESTful inference APIs using FastAPI to serve LLM predictions, vector search, and streaming responses under concurrent load. Integrated AI components with PostgreSQL, MyBatis, and application backends; documented integration contracts for cross-functional teams.
 - Secure Deployment & Production Infrastructure:** Deployed and operated AI services on AWS EC2 with NGINX and Apache Tomcat. Maintained system security, stability, and operational readiness; supported debugging, performance monitoring, and infrastructure improvements across the deployment lifecycle.
 - IoT Biometric Authentication Prototype → Production:** Took an ECG-based authentication concept end-to-end: hardware signal acquisition (ESP32, AD8232), preprocessing pipeline, CNN-LSTM model training, and classification system — demonstrating the full prototype-to-working-system delivery pattern.
- Samsung Asset Management** Gangnam, Republic of Korea
Web Developer – Dispatched Role *Sept. 2024 – Dec. 2024*
 - Enterprise Asset Management System:** Contributed to front-end and back-end development for an enterprise asset management system. Implemented UI enhancements using HTML, CSS, JavaScript, and jQuery, and supported backend functionality using Spring Boot for asset tracking and reporting workflows.
- Database and Dependable Computing Laboratory** Ajou University, Suwon, Republic of Korea
Researcher *Sept. 2017 – Apr. 2024*
 - Research-to-Implementation:** Conducted research on efficient query processing in large-scale road networks and implemented algorithmic solutions for real-world spatial data problems.
 - Large-Scale Data Processing:** Designed and evaluated distributed processing methods using Apache Spark to improve performance for large-scale spatial workloads.
 - Algorithm Design and Evaluation:** Designed algorithms, built experimental systems, analyzed results, and prepared research manuscripts for publication.
 - Blockchain and Privacy-Preserving Systems:** Designed algorithms for Hyperledger Fabric-based blockchain models to support data integrity and privacy in IoT environments.

SELECTED PROJECTS

- **ECG-Based Biometric Authentication System:** Developed a machine learning prototype using **ESP32**, **AD8232 ECG sensor**, and a hybrid **CNN-LSTM** model for user authentication from physiological signal data. Focused on sensor data acquisition, signal preparation, model design, and classification.
- **Enterprise RAG-Based Knowledge Management System:** Built an AI-powered document search and question-answering system using **LangChain**, **Ollama**, **ChromaDB**, **embeddings**, **semantic search**, and open-source LLMs. Developed document ingestion, retrieval, and inference workflows for internal technical knowledge.
- **Facial Recognition-Based Smart Fridge System:** Developed a facial recognition-based access control module for a smart fridge system. Implemented identity verification and point-based usage tracking to support personalized user access records.
- **Distributed Spatial Query Processing:** Designed and implemented distributed algorithms for spatial query processing in road networks using **Java**, **Apache Spark**, **Hadoop**, and graph-based datasets. Focused on efficient computation over large-scale network data.
- **Facility Management Backend System:** Developed backend services using **Spring Boot**, **PostgreSQL**, **MyBatis**, **AWS EC2**, RESTful APIs, optimized SQL logic, and CI/CD using **GitHub Actions**, **AWS S3**, and **CodeDeploy**.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL
- **AI & Agentic Systems:** LangChain, Ollama, RAG, agentic workflows, LLM inference, Hugging Face, prompt engineering, reranking
- **Machine Learning:** PyTorch, CNN, LSTM, time-series modeling, classification, model evaluation
- **Backend & APIs:** FastAPI, Spring Boot, RESTful APIs, streaming responses, MyBatis
- **CI/CD & DevOps:** GitHub Actions, AWS CodeDeploy, AWS S3, Docker, NGINX, Jenkins, Git
- **Cloud:** AWS EC2, GCP
- **Databases & Vector Retrieval:** PostgreSQL, MySQL, ChromaDB, Qdrant, FAISS, semantic search
- **Distributed Computing:** Apache Spark, Hadoop/YARN

RELEVANT STRENGTHS

- Strong ability to learn new domains, study technical literature, implement research ideas, and turn them into working systems. Comfortable working across data preparation, model development, backend integration, deployment, and production support. Interested in applying machine learning to real-world systems where model quality, reliability, and practical impact matter.